A study of a fictional research assistant

This document and every associated chat are fabricated for testing Adam search. No measurements or research claims are real. The study follows a fictional assistant named Cedar as it retrieves notes from a small research library.

The study asks three questions: can sparse attention reduce retrieval cost, how does latency change with context length, and which limitations should a reader notice? Search for study to find repeated matches across all four pages.

In this synthetic study, Cedar reads 120 invented notes. Each note has a title, a short summary, and a source tag. The baseline reads every note; the sparse attention variant selects a smaller candidate set.

Method and benchmark design

The benchmark contains three fictional workloads: short summaries, long context retrieval, and conflicting evidence. Every benchmark uses the same invented corpus, so the numbers below are illustrations only.

The study compares a baseline with a sparse attention variant. Retrieval recall describes how many relevant notes were found. Latency describes elapsed time before the assistant can begin an answer.

Use PDF search for benchmark or sparse attention. Use Enter to advance and Shift+Enter to go backward. The benchmark keyword also appears in multiple saved chats, so the chat history search should find more than one conversation.

Results and tradeoffs

In the fictional results, sparse attention improves median latency from 240 milliseconds to 150 milliseconds. Retrieval recall changes from 92 percent to 89 percent. This is a synthetic tradeoff, not evidence about a real model.

The study finds that shorter candidate lists can reduce latency but may omit useful evidence. A reader should inspect missed notes before accepting an apparently faster system.

The study repeats the benchmark at three context lengths. The latency advantage grows in this fabricated example, while recall remains sensitive to the candidate selection rule. No confidence interval should be treated as a real statistical result.

Fictional variant	Latency	Recall
Baseline	240 ms	92%
Sparse attention	150 ms	89%

Limitations and search checklist

This study has deliberate limitations: an invented corpus, a small sample, and no independently measured results. It exists to make PDF search and chat search easy to inspect.

Try study in PDF search. Type s, st, stu, stud, and study slowly, then backspace. Only the current query should stay highlighted. Search for latency and move through results on different pages.

Open Chat history and search benchmark to find several conversations. Search pineapple to find a word that appears only in a chat message. Search Moonstone to find a title-only match. Open a result to jump into that chat.